

Andrus, R. A., Hicke, J. A., & Meddens, A. J. H. (2025). Spatiotemporal characteristics of tree mortality from bark beetle outbreaks vary within and among bark beetle-host tree associations in the western United States. *Forest Ecology and Management*, 576, 122382. <https://doi.org/10.1016/j.foreco.2024.122382>

This paper looks specifically at 9 of the most damaging bark beetle-host tree associations with a spatiotemporal lens. The results found notable and significant findings in the mortality area of associated hosts that have a maximum annual mortality area as well as a cumulative mortality area. This is useful to our research because it highlights beetle preferences and speaks about how damaging the beetles can be.

Assal, T. J., Sibold, J., & Reich, R. (2014). Modeling a Historical Mountain Pine Beetle Outbreak Using Landsat MSS and Multiple Lines of Evidence. *Remote Sensing of Environment*, 155, 275–288. <https://doi.org/10.1016/j.rse.2014.09.002>

This paper incorporates more intensive modeling techniques with spatial analysis and data which will be a really good reference point for our project. They build a linear model based off of historical data about beetle kill in California and then can project it back onto a map which will be really helpful to us when we are designing models with our spatial data's attribute tables.

Bentz, B. J., & Endreson, D. (n.d.). *Evaluating satellite imagery for estimating mountain pine beetle-caused lodgepole pine mortality: Current status.*

Bode, E. T., Lawrence, R. L., & Powell, S. L. (2018). Time-series approach for mapping mountain pine beetle infestation extent and severity in the U.S. Central Rocky Mountains. *Journal of Applied Remote Sensing*, 12(04), 1. <https://doi.org/10.1117/1.JRS.12.046030>

Using linear models, time series, and random forest prediction this study will lend a great help to ours with their very similar methods of tracking pine beetle outbreak severity. After training a model on a set of sample data to determine from aerial imagery where beetle kill were over the years they could use time series analysis to show change over the years with a raster map projected back onto the site being studied.

Chapman, T. B., Veblen, T. T., & Schoennagel, T. (2012). Spatiotemporal patterns of mountain pine beetle activity in the southern Rocky Mountains. *Ecology*, 93(10), 2175–2185. <https://doi.org/10.1890/11-1055.1>

This paper used spatiotemporal patterns looking at ponderosa and lodgepole pines from 1996-2010 in parts of Colorado and Wyoming. The findings of this paper say that warm temperatures and low precipitation are the favorable conditions for pine beetles, which are findings that support ours.

Cooke, B. J. (n.d.). *Forecasting mountain pine beetle-overwintering mortality in a variable environment*.

This paper talks about overwintering mortality in pine beetles, with data collected in Alberta, Canada. The researchers were able to successfully predict pine beetle mortality, and following some of their methods could be advantageous for us.

Kamarianakis, Y. (n.d.). *SPATIAL-TIME SERIES MODELING: A REVIEW OF THE PROPOSED METHODOLOGIES*.

Liang, L., Chen, Y., Hawbaker, T. J., Zhu, Z., & Gong, P. (2014). Mapping Mountain Pine Beetle Mortality through Growth Trend Analysis of Time-Series Landsat Data. *Remote Sensing*, 6(6), 5696–5716. <https://doi.org/10.3390/rs6065696>

These researchers made maps that covered disturbance of pine beetles, and used them to predict pine beetle movement. They were able to make highly accurate predictions, and looking into what features they used could be very beneficial to us

Liu, Y., & Estep, D. (n.d.). A multiscale stochastic cellular automata model for dispersion of mountain pine beetles. *Royal Society Open Science*, 12(6), 241934. <https://doi.org/10.1098/rsos.241934>

The authors of this paper created a very unique and in depth model to track biotic spread through an area which they implemented to specifically follow pine beetle spread in parts of Canada. Though this model is far too complex for our project and requires an abundance of data we don't have, we can pull aspects of their model and use assumptions they made in the models we build.

Logan, J. A., & Bentz, B. J. (1999). Model Analysis of Mountain Pine Beetle (Coleoptera: Scolytidae) Seasonality. *Environmental Entomology*, 28(6), 924–934.

<https://doi.org/10.1093/ee/28.6.924>

This paper goes over how temperature affects pine beetles, specifically with seasonal temperature changes. They found that pine beetles tend to adapt to seasonal temperature changes quite well, but the more unstable climate change temperatures can impact them. This is a factor that will be important to think about, and reviewing their processes and analysis could be very useful to us.

Meddens, A. J. H., & Hicke, J. A. (2014). Spatial and temporal patterns of Landsat-based detection of tree mortality caused by a mountain pine beetle outbreak in Colorado, USA. *Forest Ecology and Management*, 322, 78–88.

<https://doi.org/10.1016/j.foreco.2014.02.037>

This paper approaches modeling pine beetles through temporal and spatial data with a GAM model and time series analysis. They, contrary to us, are trying to predict tree mortality rather than outbreaks but it's possible that we can shift some of these concepts over to better suit our means.

Negrón, J. F. (2020). Within-Stand Distribution of Tree Mortality Caused by Mountain Pine Beetle, *Dendroctonus ponderosae* Hopkins. *Insects*, 11(2), 112.

<https://doi.org/10.3390/insects11020112>

This paper looks more specifically at ponderosa pines and pine beetles preferences for tree stands. The study finds that the pine beetles prefer denser stands as well as denser forests. This study also mentions that thinning forests (which is one of our solutions) is a viable solution for pine beetles so long as those going through the restoration efforts are mindful of how the beetles are responding so as to not make the issue worse.

Shahriari, S., Sisson, S. A., & Rashidi, T. (2025). Modelling time series with temporal and spatial correlations in transport planning using hierarchical ARIMA-copula Model: A Bayesian approach. *Expert Systems with Applications*, 274, 126977.

<https://doi.org/10.1016/j.eswa.2025.126977>

This paper goes over modeling time series data and predictive modeling. This will be very informative to us on how to actually create predictive models, which is one of our goals.

Wulder, M. A., Dymond, C. C., White, J. C., Leckie, D. G., & Carroll, A. L. (2006). Surveying mountain pine beetle damage of forests: A review of remote sensing opportunities. *Forest Ecology and Management*, 221(1), 27–41.

<https://doi.org/10.1016/j.foreco.2005.09.021>

This paper talks about new developments in remote sensing and how to use remote sensing to survey mountain pine beetle activity. This paper talks about new technologies which may be interesting to reference as well as future driven approaches to pine beetles.

Yumpu.com. (n.d.). *Using GIS to Model the Risk of Mountain Pine Beetle Infestation in ...*

Yumpu.Com. Retrieved March 2, 2026, from

<https://www.yumpu.com/en/document/view/26841520/using-gis-to-model-the-risk-of-mountain-pine-beetle-infestation-in->

Due to the older nature of this paper the spatial analysis methods are very proprietary which lends itself well to our project when we look for correlations between different variables in our spatial dataset and visually represent it using a GIS. We can use methods that they did such as weighting contextual datasets like tree density in order to model how pine beetles spread based on natural characteristics.